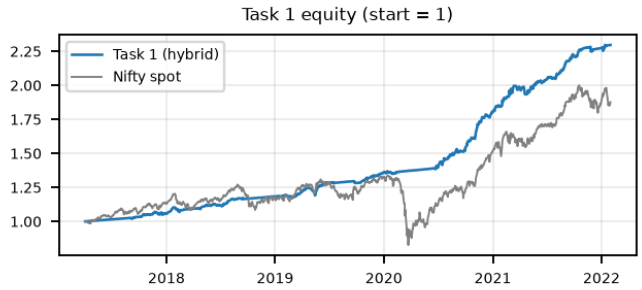
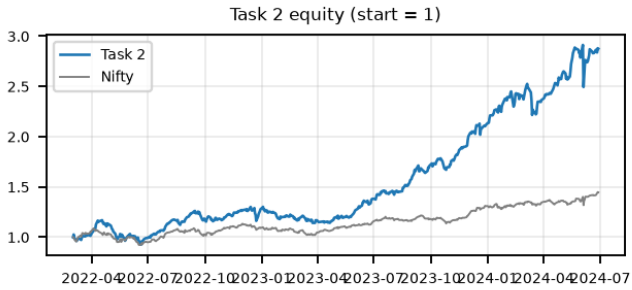
Quant Research take-home: Nifty timing (Task 1) and NSE momentum (Task 2)

Harness results after costs, visible data only. Parameters were tuned on the same data, so results are in-sample; hidden windows are not shown.

	Sharpe	Max DD	CAGR	Trades	Bar to beat
Task 1: Apr 2017-Jan 2022	2.17	-4.7%	18.8%	468	Sharpe > 1.0 and DD better than -15% (internal: 1.22, -12.8%, 23.3%)
Task 2: Mar 2022-Jun 2024	2.10	-20.9%	57.5%	176	Sharpe > 1.9 and DD better than -25% (internal: 1.93, -23.9%, 51.6%)





1. Task 1: Nifty futures, overnight continuation

Rule. Long overnight when the prior close is above its 100-day average and the index is up on the day at the decision bar. Decision at the close of the 14:30 bar (timestamp 14:45), fill at the 14:45 open; held through the 14:45, 15:00 and 15:15 bars, the overnight gap and the 09:15 bar; flat from the 09:30 open (p = 2, q = 1). Otherwise cash (6%). **Source of return.** Mean overnight gap +15.6 bp (t = 7.5, positive in every full year); session drift about -24% a year; 09:15 bar -6.5 bp. The unfiltered overnight version scores Sharpe 0.35; invested nights net +11.8 bp against -4.6 bp otherwise (Welch t = 3.7; none of 5000 random night sets of the same size matches). Entering p bars before the close keeps the edge under a 1 to 2 bar delay; a 3-bar delay gives Sharpe -2.4. **Limits.** In market 6% of bars (reference 55%). CAGR 18.8% includes 6.06% cash (12.0% above cash) and is below the reference 23.3%. The edge is the spot-index overnight gap; futures executability is untested and persistence after 2022 is unknown.

Three approaches, same harness

Approach	Sharpe	Max DD	CAGR	Trades	In mkt	Delay 1	Delay 2	2x cost: Sh / DD	No cost
A: pure overnight gap (last bar + gap, SMA100)	3.01	-6.5%	27.1%	827	3%	-4.86	-4.43	1.54 / -7.1%	4.45
B: daily trend long/flat (SMA50, 20d vol<35%)	0.86	-12.5%	15.5%	37	69%	0.72	0.71	0.79 / -13.0%	0.93
C: hybrid (submitted)	2.17	-4.7%	18.8%	468	6%	2.19	1.88	1.23 / -5.2%	3.08

A has the highest Sharpe but is negative at a 1-bar delay. B is robust but peaks at Sharpe 0.86 (best of about 30 daily trend variants). C holds Sharpe above 1.8 at 1 to 2 bar delay and 1.23 at double costs.

Parameters (4): sma 50-200 (100); p 1-3 (2); q 1-3 (1); mom -0.2% to +0.2% (0). Dropped from the code: volatility cap, rolling kill-switch. Sensitivity, Sharpe (1-bar-delay Sharpe, max DD):

sma \ mom	-0.002	0.0	0.002
50	2.32 (d1 2.32, DD -3%)	2.40 (d1 2.35, DD -3%)	2.34 (d1 2.31, DD -2%)
100	2.09 (d1 2.13, DD -5%)	2.17 (d1 2.19, DD -5%)	2.16 (d1 2.05, DD -4%)
200	2.04 (d1 2.05, DD -5%)	2.11 (d1 2.11, DD -5%)	2.15 (d1 2.03, DD -4%)

p \ q	1	2	3
1	2.75 (d1 2.17, DD -2%)	2.50 (d1 2.06, DD -3%)	2.39 (d1 2.03, DD -3%)
2	2.17 (d1 2.19, DD -5%)	1.92 (d1 2.09, DD -5%)	1.83 (d1 2.07, DD -5%)
3	2.07 (d1 1.92, DD -3%)	1.90 (d1 1.82, DD -4%)	1.80 (d1 1.78, DD -4%)

18 cells: Sharpe 1.80-2.75 (1-bar delay 1.78-2.35), max DD no worse than -5%. p = 1 reaches 2.75 but is -2.5 at a 2-bar delay. p and sma were set from an earlier scan on the same data.

Year-wise	2017	2018	2019	2020	2021	2022	Sub-period Sharpe (DD)
Return	5.7%	11.9%	14.7%	32.7%	26.2%	1.0%	2017-18: 1.10 (-3%) 2019: 1.71 (-5%) 2020: 3.52 (-2%) 2021: 2.46 (-3%)

Best year removed (2020, +33%): Sharpe 1.74 (full 2.17); any single year removed: 1.74-2.39. 2022 is 22 days. Stationary-bootstrap 95% CI of the Sharpe: 1.2 to 3.1. Deflated Sharpe: probability 0.999 over the 108 settings of this family, 0.24 pooling all structures tried. **Turnover and costs:** sum of |position changes| 194 per year, about 97 trades a year; Sharpe 3.08 before costs, 2.17 after; 1.23 at 2x costs, 0.25 at 3x, -0.74 at 4x. **Split-sample check** (72 settings, picked on one half by mean of base and 1-bar-delay Sharpe, tested on the other; design set after seeing both halves; setting ranks correlate 0.02 across halves; every setting is better in the second half): picked on Apr 2017-Aug 2019, tested Sep 2019-Jan 2022: Sharpe 2.76 (DD -2.8%); picked on the second half, tested on the first: 1.38 (DD -1.6%). Median setting Sharpe 2.92 in the second half vs 1.11 in the first. Submitted setting: 1.17 first half, 2.92 second.

Failure modes. Overnight premium disappears or reverses (the strategy then earns cash minus costs on about 97 round trips a year); futures open differently from the spot gap; strong days reverse overnight; costs near 3x the stated level; special sessions with unusual bar times can trigger an unintended 1-bar trade.

2. Task 2: NSE momentum, volatility-capped

Rule. Rank liquid stocks by 6-month return; hold the top 10 equally weighted among names above their 50-day average and at or below the 75th percentile of 60-day volatility; rebalance every 21 days. Liquidity floor Rs 10 cr; stocks with a one-day drop over 40% in the past year are excluded.
Parameters (5): n 8-12 (10), lb 84-168 (126), rebal 10-30 (21), sma 40-70 (50), volq 0.60-1.00 (0.75). The liquidity floor was also tried at Rs 25 cr, a sixth tuned value. **History.** The unrestricted version was a high-volatility book (median volatility 54% vs 33% for the universe). The cap kept the full-window Sharpe and lifted the Sharpe with the best 15% of names removed from 0.31 to 1.25 at 0.4% costs; that is a hindsight stress, not a survivorship estimate, and the cap was chosen partly against it. A first version used a breadth brake (Sharpe 2.04, DD -18.1%) and was replaced after failing three checks set in advance (failed_ideas.md). There is no market-level brake and the sample has no long bear market. About 90 configurations (about 200 harness runs) were scored on this window.

Metric	Value	Metric	Value
Sharpe / Max DD / CAGR	2.10 / -20.9% / 57.5%	Year-wise	2022: 28% 2023: 65% 2024: 36% (2024 = H1)

Trades / win rate / avg hold	176 / 57% / 32 d	Turnover	15.4x per year (one-way); time in market 100%
Cost drag (per side)	0%: Sharpe 2.23, CAGR 62% | 0.2%: 2.10, 58% | 0.4%: 1.97, 53% | 0.6%: 1.84, 49%	Best year removed	Sharpe w/o 2022: 2.48, w/o 2023: 1.72, w/o 2024: 2.14 (full 2.10)
Top 10 trades (exact sizes)	36% of gross profit, 51% of closed net P&L;	Sub-windows (Sharpe)	2022: 1.40, 2023: 2.78, 2024 H1: 1.43

volq \ sma (Sharpe / DD / CAGR)	**40**	**50**	**70**
0.6	1.59 / -21% / 33%	1.57 / -20% / 32%	1.50 / -24% / 31%
0.75	2.16 / -18% / 59%	2.10 / -21% / 58%	1.88 / -21% / 50%
0.9	1.73 / -24% / 51%	1.74 / -24% / 52%	1.74 / -24% / 52%

n \ lb (Sharpe / DD / CAGR)	**84**	**126**	**168**
8	2.20 / -20% / 60%	2.01 / -21% / 57%	2.28 / -21% / 65%
10	2.54 / -18% / 68%	2.10 / -21% / 58%	2.18 / -24% / 58%
12	2.52 / -16% / 67%	2.24 / -18% / 60%	2.16 / -24% / 55%

rebal \ volq (Sharpe / DD / CAGR)	**0.6**	**0.75**	**0.9**
10	1.71 / -22% / 37%	2.31 / -15% / 64%	2.14 / -22% / 68%
21	1.57 / -20% / 32%	2.10 / -21% / 58%	1.74 / -24% / 52%
30	1.71 / -18% / 37%	1.97 / -17% / 50%	2.03 / -25% / 63%

Sensitivity. n x lb is flat (Sharpe 2.0-2.5); volq is not: 0.75 is a peak (0.60: 1.5-1.7; 0.90: 1.7-2.1). **Calendar.** Shifting the rebalance start offset (0 to 20 days) gives Sharpe 1.26 to 2.33 (mean 1.76); the submitted offset ranks 2nd of 21. A staggered-tranche variant gives 1.60-2.02 (mean 1.81) and missed its range rule (0.42 vs 0.3); not adopted. **Universe.** Random 30% of names removed (6 seeds): mean Sharpe 2.17 (min 1.94). Best 15% removed with hindsight: 1.40 (CAGR 30%).
Failure modes. Momentum crash (no brake; exits come only from the trend filter and the rebalance); sideways market with repeated leader reversals at 0.4% round trip; the cap excluding leaders of a new regime; liquidity below the Rs 10 cr floor; data jumps outside the guard (one-day drops over 40% only). Cash earns 0%; the book is almost always fully invested.

3. Benchmarks and stress

Task 2 benchmark (same window)	**Sharpe**	**CAGR**	**Max DD**
Strategy (0.2%/side)	2.10	58%	-21%
Equal-weight of all liquid stocks in the file (daily rebalanced, no costs, survivor-biased)	2.09	39%	-20%
10 random liquid stocks, rebalanced every 21 days, same costs (12 seeds)	1.52 (1.12 to 1.93)	32%	-24% worst
Nifty 50	1.26	17%	-15%

Task 2. Equal-weight basket of the liquid stocks in the file: Sharpe 2.09 against 2.10 for the strategy; random 10-stock books: about 1.5. The list is survivor-biased and the window is a bull market. Against the basket: CAGR +18 points, information ratio 1.01, alpha 9% a year, beta 1.13, over 2.3 years. With a Rs 25 cr liquidity floor (unrestricted version) the edge over the basket nearly disappears. Working range on new data: about 0.8-1.6; negative in a momentum crash.

Task 1 scenario (submitted setting)	**Sharpe**	**CAGR**	**Max DD**
As reported	2.17	19%	-4.7%
Delay 1	2.19	19%	-3.5%
Delay 2	1.88	17%	-3.7%
2x costs	1.23	13%	-5.2%
3x costs	0.25	7%	-5.6%
Delay 1 + 2x costs (realistic)	1.25	13%	-3.7%
Delay 2 + 3x costs (harsh)	-0.07	6%	-5.4%

Task 1. Delay 1 plus 2x costs: Sharpe 1.25 (CAGR 13%); median of 54 nearby settings 1.16, 81% above 1.0. Delay 2 plus 3x costs: no edge (median 0.04). Gap +15.7 bp a night against about 5 bp round-trip cost; costs take 23% and carry 9% of gross profit; 2020 is 41% of total P&L.; The first bar of the day is negative after both up and down gaps (-7.2 bp / -4.6 bp). Working range: about 1.0-1.5. Futures 1-minute data and the official index close would test executability.

4. Survivorship bias (Task 2 universe)

The universe is the set of stocks liquid in October 2026; names liquid in 2022 that later fell, merged or were delisted are missing (median stock +98% in the window vs +44.6% for the Nifty). Tests, none a survivorship estimate: equal-weight basket (Sharpe 2.09); best 15% of names removed with hindsight (Sharpe 1.40; worst 15% removed gives 2.20, random 15% gives 2.24); random 30% removal (2.17); 3% of trades replaced by -60% (median Sharpe 1.46, additive approximation). Proper estimate: a point-in-time universe including delisted names, same code on both. Hidden window: the selection date is its end, so the look-ahead horizon is up to 2.3 years (4.6 for the visible window); the bias is smaller, not absent. See survivorship_note.md.